STAT 361 Assignment 2 Question 5

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a) Finding Least Squares Estimator $\hat{\beta}$

First, we set up the normal equation for this problem.

$$S(\beta) = \sum_{i=1}^{n} [Y_i - \beta x_i]^2$$
$$\frac{\partial S}{\partial \beta} = -2 \sum_{i=1}^{n} [Y_i - \beta x_i] x_i$$

Set this derivative equal to zero, solve for β

$$\begin{aligned}
0 &= -2 \sum_{i=1}^{n} [Y_i - \beta x_i] x_i \\
\Rightarrow 0 &= \sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} \beta x_i^2 && \textbf{(divide by } -2\textbf{)} \\
\Rightarrow \beta \sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} Y_i x_i && \textbf{(pull out constant } \beta\textbf{)} \\
\Rightarrow \beta &= \frac{\sum_{i=1}^{n} Y_i x_i}{\sum_{i=1}^{n} x_i^2} \\
&= \hat{\beta}
\end{aligned}$$

Where $\hat{\beta}$ is the least squares estimator of β.

(Answer 5a)

b) Unbiasedness and Variance Comparison of $\hat{\beta}, \tilde{\beta}$

Similar to the assumptions in the slides, let all Y_i be independent, $Var(Y_i) = \sigma^2$ (common variance), and $E[Y_i] = \beta x_i$ for the linear model in a). We treat x_i as a constant. Thus,

$$\begin{aligned} E[\hat{\beta}] &= \frac{\sum_{i=1}^n x_i E[Y_i]}{\sum_{i=1}^n x_i^2} \\ &= \frac{\sum_{i=1}^n \beta x_i^2}{\sum_{i=1}^n x_i^2} \qquad \textbf{(}\beta \textbf{ is a constant)} \\ &= \beta \end{aligned}$$

$\therefore E[\hat{\beta}]$ is unbiased

$$\begin{aligned} E[\tilde{\beta}] &= \frac{\sum_{i=1}^n E[Y_i]}{\sum_{i=1}^n x_i} \\ &= \frac{\sum_{i=1}^n \beta x_i}{\sum_{i=1}^n x_i} \qquad \textbf{(}\beta \textbf{ is a constant)} \\ &= \beta \end{aligned}$$

$\therefore E[\tilde{\beta}]$ is unbiased.

Now for deriving the variance:

$$\begin{aligned} Var(\hat{\beta}) &= \frac{Var\left(\sum_{i=1}^n x_i Y_i\right)}{Var\left(\sum_{i=1}^n x_i^2\right)^2} \\ &= \frac{\sum_{i=1}^n x_i^2 Var(Y_i)}{\left(\sum_{i=1}^n x_i^2\right)^2} \\ &= \frac{\sum_{i=1}^n x_i^2 \sigma^2}{\left(\sum_{i=1}^n x_i^2\right)^2} \\ &= \frac{\sigma^2}{\sum_{i=1}^n x_i^2} \\ &= \frac{\sigma^2}{n\overline{x}^2} \qquad \textbf{(property from slides)} \end{aligned}$$

$$\begin{aligned} Var(\tilde{\beta}) &= \frac{Var\left(\sum_{i=1}^n Y_i\right)}{Var\left(\sum_{i=1}^n x_i\right)} \\ &= \frac{\sum_{i=1}^n Var(Y_i)}{\sum_{i=1}^n x_i^2} \\ &= \frac{n\sigma^2}{\sum_{i=1}^n x_i^2} \\ &= \frac{n\sigma^2}{n^2\overline{x}^2} \qquad \textbf{(property from slides)} \\ &= \frac{\sigma^2}{n\overline{x}^2} \end{aligned}$$

No need to use the hint, we may directly compare the variances:

$$\begin{aligned} \frac{\sigma^2}{n^2\overline{x}^2} &\leq \frac{\sigma^2}{n\overline{x}^2} \\ \Rightarrow Var(\hat{\beta}) &\leq Var(\tilde{\beta}) \end{aligned}$$

This is true as the denominator has n to a greater power in $Var(\hat{\beta})$, and the two variances may be equal when $n = 1$, or when x_i are equal, thus $x_i = \overline{x}\ \forall i$.
This is also intuitive because $\tilde{\beta}$ has less terms in its least squares estimator relative to $\hat{\beta}$; the more variables the higher the variance of a model typically.

(Answer 5b)

c) Use of distribution assumptions

No, we did not use distribution assumptions throughout this question at all. We only used the assumptions that:

- $E[\epsilon_i] = 0$ to show unbiasedness
- $Var(\epsilon_i) = \sigma^2$ to then also assume $Var(Y_i) = \sigma^2$ by independence

This follows with the similar derivations present in the slides, where we didn't show ϵ_i at all for brevity.

(Answer 5c)